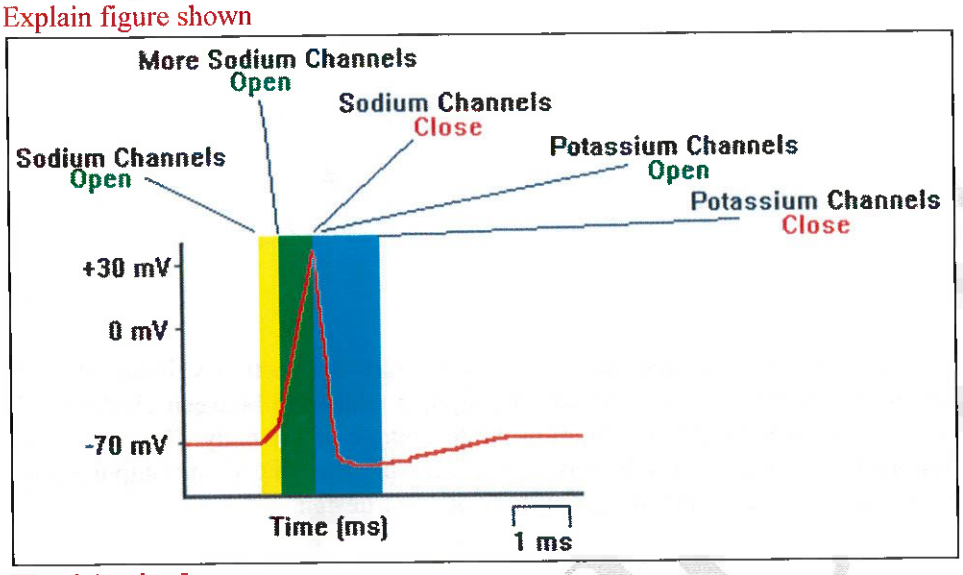
1. This question relates to the design of a multichannel instrument for *measuring bio-potentials from the muscles.*

a) Explain how a neural action potential is generated referring to the ionic flow of sodium and potassium ions. You are required to sketch the waveform of an action potential and annotate each phase, explaining the mechanism contributing to these. [4]

[bookwork]

Explain figure shown



[Total 4 points]

b) Table 1.1 lists five commonly observed bio-potentials and their amplitude and frequency characteristics. A doctor requires an instrumentation system to record from a set of electrodes connected to the back muscles shown in Figure 1.1.

Design a complete instrumentation system for recoding bio-potentials from the back according to the following specifications:

Instrumentation Amplifier Gain: 10
Low-Pass Filter Roll off: 40dB/decade
High-Pass Filter Roll off: 60dB/decade
Supply Range: -3V to 3V

State all assumptions and values of resistors and capacitors used. You may assume that the instrumentation system utilizes the whole dynamic range available and 10nF capacitors are available for the filters.

Using table below for EMG:
High pass cut off: f_{HP}=20Hz
$R_{HP}=1/2\pi Cf_{HP}$ = 1/2π(10nF)(20Hz) =7.95 MΩ [1 point]
Low pass cut off: f_{LP}=2KHz
$R_{LP}=1/2\pi Cf_{LP}$ = 1/2π(10nF)(2KHz)=79.5KΩ [1 point]

Low pass 40dB/decade: Sallen Key can be used
High pass 60dB/decade: Opamp with RC + Sallen Key [1 point]

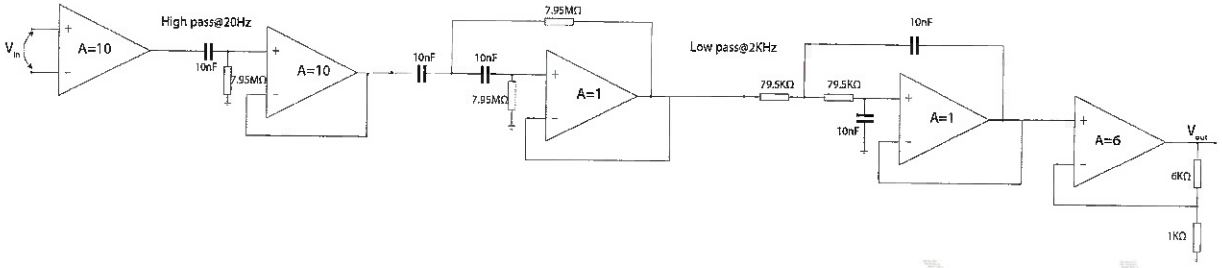
Instrumentation amplifier gain:
$V_{OUT} = -(V_1 - V_2)(1 + 2R_2/R_1)(R_4/R_3)$
For gain of 10 we select R4=R3=R2= 100 KΩ and R1= 10 KΩ [1 point]

Supply Range: -3V to 3V
VDD=3V, GND=0V, VSS=3V
Max EMG = 100mV (from table)

V_{inptp} x A1 x A2=$6V_{ptp}$
A1=10
100mVx10xA2=6V
A2= 6 (gain of filter and final stage) [1 point]
Realised with a non-inverting amplifier with R1=6KΩ and R2=1KΩ [1 point]

A potential solution [4 points]:

[10]



c) The function of the system is to measure the muscle fibre conduction velocity (MFCV), determined by the propagation delay between the signals measured between electrodes 1-2 and 3-4. Electrode 5 is used to set the common mode voltage of the body. Design a system using your front end designed in 1.b which is able to measure the MFCV and output a digital number representing this value. State all assumptions in your design.

[6]

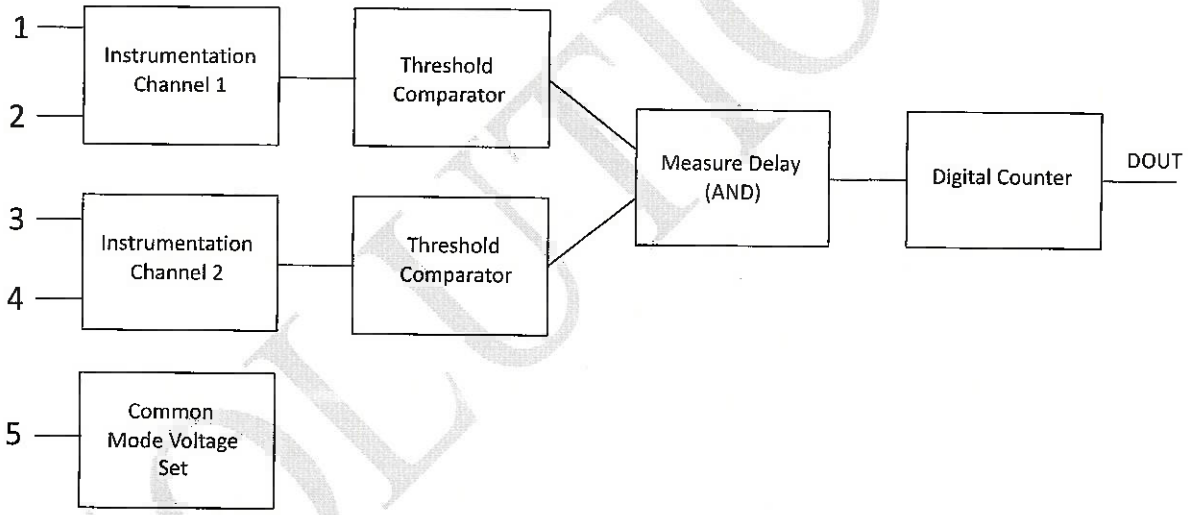


Table 1.1

Signal	**Frequency range (Hz)**	**Amplitude range (mV)**
ECG	0.1 – 300	0.05 – 3
EEG	0.5 – 40	0.001 – 1
EMG	20 – 2000	0.001 – 100
Neural Spikes	300-5000	0.001– 0.5
Local Field Potentials	10-200	0.001 – 5

SOLUTION

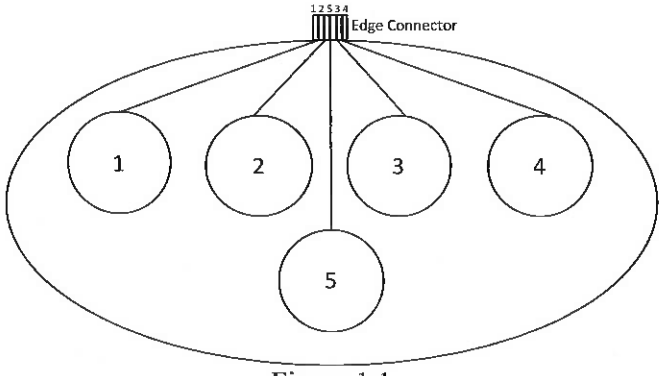


Figure 1.1

2. This question relates to *implantable neural prostheses for restoring vision.*

d) Briefly discuss the relative advantages and disadvantages of an *implantable neural prosthesis for restoring vision* that interfaces to the: (i) retina; (ii) optic nerve; (iii) visual cortex. [6]

(i) Retinal interfacing only applicable to cases of blindness where the output of retina (e.g. bipolar/ganglion cells), optic nerve and cortex are all intact. For example, in RP or AMD. This is a small portion of the total blind population. [1] A desirable feature is that the retina is one of the few parts of the brain that are relatively easily accessible (via the eye). So much less invasive than a cortical implant for example. [1]

(ii) Optic nerve can be interfaced even when the retina has been completely damaged (for example in late stage AMD or due to diabetic retinopathy). [1] However, the main challenge is the optic nerve has some 1M fibers (Ganglion axons) cannot be individual addressed. Any electrode interface has to be invasive to the nerve (penetrating) thus potentially damaging. [1] Cuff electrodes on the other hand would provide no selectivity.

(iii) Visual cortex would be the most invasive option (requiring brain surgery) but would be applicable to people with a damaged retina and/or optic nerve (for example, certain types of tumours form around nerves requiring them to be removed thus blinding the patient). [1] Spatial mapping in the visual cortex would be more challenging than in retina (not perfectly 2D focal plane as in retina). [1]

e) Given a desired resolution of 16384 pixels (i.e. 128 x 128) is required for a retinal prosthesis, determine:

i. The area per electrode in contact with tissue (hint: you will need to estimate the total area of retina to be targeted); [2]

Assuming the active part of the retina covers a 5mm diameter disc -> corresponds to a square focal plane of approx. 3mm x 3mm. So to have 128 x 128 pixels would require each pixel to be at 3mm/128 = 23 micron pitch (centre-to-centre). Assuming this is arranged to have a 12 x 12 micron electrode and 11 micron separation would give an electrode area of 144 sq micron => $1.44 \times 10^{-6} cm^2$.

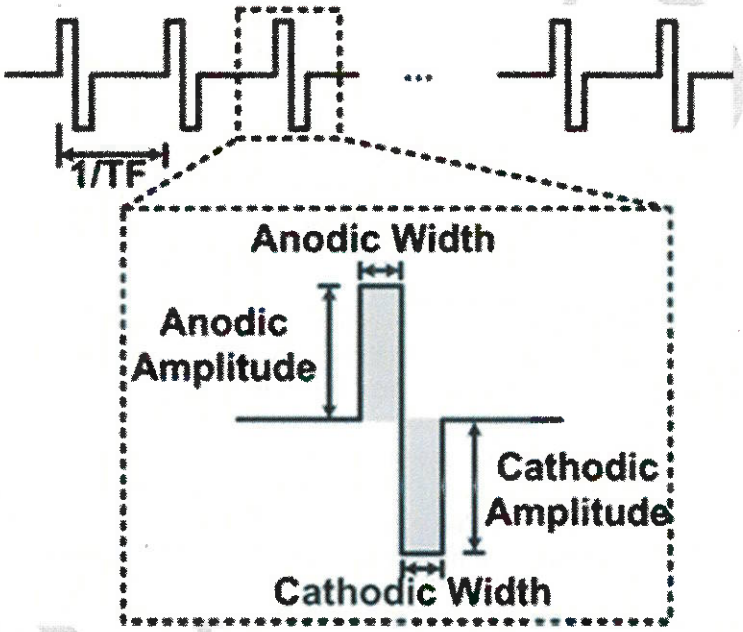
ii. Given the following material properties, select an electrode coating, stating your rationale; [2]

Material	Charge Density
Stainless steel	50μC/cm^2
Platinum Iridium	150μC/cm^2
Iridium Oxide	2.5mC/cm^2

Given the electrodes are very small and thus to maximize allowable stimulation current, as a retinal prosthesis would have a very large number of electrodesand thus TDM would constrain stimulation cycle time [1] select material with highest charge density -> thus Iridium Oxide (2.5mC/cm2) would be the preferable. [1]

iii. For a *current-mode, anodic-first biphasic stimulation profile*, estimate the maximum allowable stimulation current, and sketch the waveform profile; [4]

Assuming for useful vision, a minimum of 20 frames/s at resolution of 128x128 pixels, and not being able to do concurrent stimulation (i.e. only one pixel at a time). This would mean the stimulator would essentially have to raster scan [1] the retinal tissue resulting in a stimulation cycle of 1/16384*20 = 3us per pixel. [1] So given the electrode size is 1.44x10-6cm2 and charge density=2.5mC/cm2 -> maximum charge capacity would be 3.6nC. [1] If this needs to be delivered in 1.5us (1.5us per phase x 2 = 3us cycle time) => 2.4mA stimulation current amplitude. [1]



Anodic width = cathodic width = 1.5us
Anodic amplitude = cathodic amplitude = 2.4mA maximum allowable
TF= repetition time = 1/20Hz = 50ms

iv. Estimate the total power consumption stating any assumptions (e.g. frame rate, resolution, stimulation protocol, etc). [2]

Since electrodes are very small – assume a very large electrode impedance = 100kOhm. So to achieve a 2.4mA stimulation current would need 2.4mA x 100kOhm = 240V!!!! If this was a feasible voltage to work with (too high), required power = 240V x 2.4mA x 50% (average current amplitude) = 288mW. [1] This includes only the stimulation power requirement and assumes power delivery and stimulus generation is 100% efficient. If the stimulation power is 25% the total power consumption -> overall power for the implant would be ~1W [1]

f) Suppose that after the implantation it was established that the charge stimulus required to achieve neural activation is 10 times higher than what is achievable for the implanted

electrode area and selected material, how can this be addressed (without changing the electrode material or size). [4]

One possibility would be to compromise on the frame rate – perhaps reducing this to 5 frames/s or 10 frames/s such as to increase the maximum allowable stimulation cycle [1]. Another approach would be to develop an advanced stimulation strategy such as what is used in cochlear prosthetics. In the answer given above we have considered a continuous interleave sampling (CIS) type strategy – to raster scan all pixels irrespective of whether they are active (have illumination) or not [1]. One alternative could be to select only the brighter pixels and selectively stimulate only pixels that have higher illumination. This would reduce the total number of pixels being stimulated and thus increase the available stimulation time per pixel [1]. This would be comparable to some of the peak stimulation strategies used in CIs. Another alternative is to do some simple image processing on the captured image (e.g. edge detection) and stimulate only those pixels. [1]

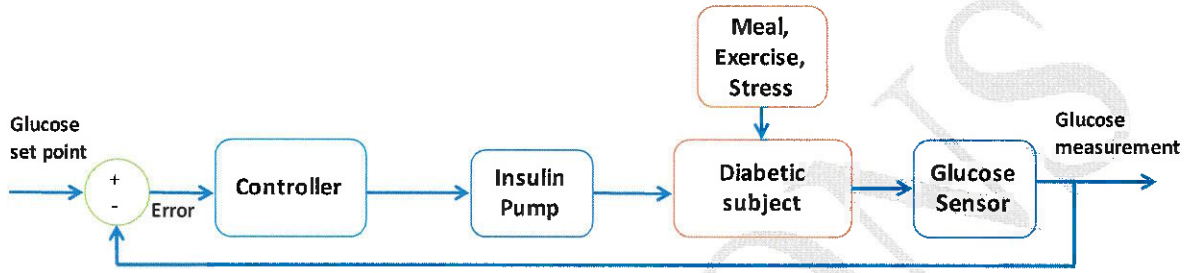
SOLUTIONS

3. Figure 3.1 shows an artificial pancreas system for control of blood glucose in diabetes. The system uses a continuous glucose sensor to measure a physiological glucose range of 0-20mM from interstitial fluid. The Glucose sensor comprises a working electrode (WE) with an area of 1mm^2, a reference electrode (RE) (area:0.5mm^2) and a counter electrode (CE) (area:2mm^2). The instrumentation is implemented in CMOS and operates on a single supply with a VDD of 1.8V.

   a) Sketch a top-level diagram of the closed-loop system for this artificial pancreas and describe its operation, identifying the devices in Figure 3.1 and also highlighting three main disturbances to the system.

Artificial Pancreas consists of continuous glucose sensor, insulin pump and controller.

Closed-loop diagram:



[2]

Explanation of operation (bookwork) highlighting Three disturbances: Meal, Exercise and Stress. [3]

[5]

The working electrode of the glucose sensor is coated with glucose oxidase to give a peak current to glucose at V_{Cell}=0.7V according to the following reaction:

$$\text{Glucose}+O_2 \xrightarrow{Glucose\ Oxidase} \text{gluconic acid}+ H_2O_2$$

$$H_2O_2+O_2 \xrightarrow{V=0.7V} 2H^++2e^-$$

   b) Calculate the peak redox current for the reaction above using the Randels Equation below. You may assume the number of electrons n is solely given by the oxidation of H_2O_2, the scan rate is 1V/sec and the diffusion coefficient D=4.38X10^{-8} m^2/s:

$$i_p = 2.69x10^5 \cdot n^{3/2} \cdot A \cdot D^{1/2} \cdot C \cdot v^{1/2}$$

n=2
A=1mm2
D=4.38X10^{-8} m2/s
C=20mM
V=1V/sec
I= 2.69*10^5*(2)$^{1.5}$*(10^{-6})*(4.38X10^{-8})$^{0.5}$*(10^{-3})*20*(1)$^{0.5}$= 3180 nA

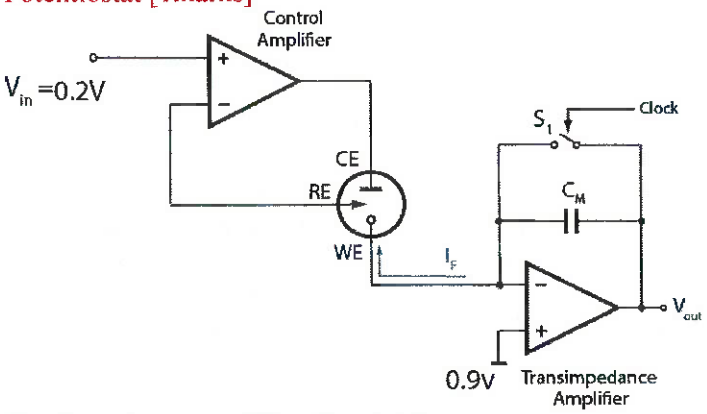
[Total 2 points]

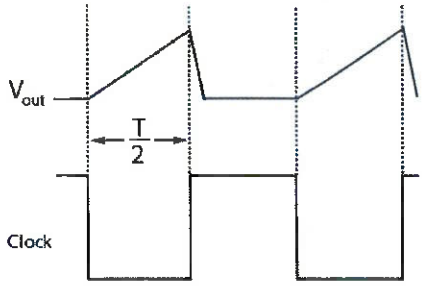
[2]

   c) Design a potentiostat and trans-impedance amplifier (TIA) to instrument to the sensor making full use of the dynamic range available. Annotate the schematic and specify all component values. You may assume a 32KHz ring oscillator exists on the device to provide a digital clock.

Calculation of capacitor value to maximise dynamic range:
Vout max = 900mV since common mode is at 0.9V for 1.8V supply
IF=CΔV/ΔT/2 =$C_m V_{outMax}$ *2f= C_m0.9V *64KHz =3180nA
Cm=55.2pF
[Total 3 points]

Potentiostat [1marks]



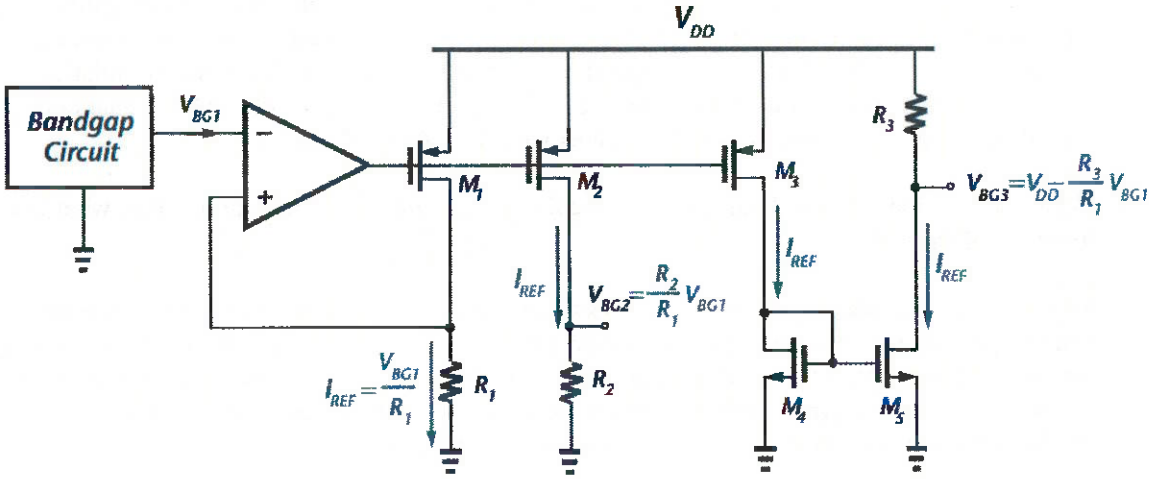


Tansimpedance amplifier [2 points]
Correct voltages [2 points]

[8]

d) Sketch a suitable circuit to provide a stable voltage to your potentiostat given that you are provided with a stable 1.2V Bandgap reference circuit on chip. You are required to calculate suitable resistor values for your voltage reference.

VBG1=1.2.
Choose Iref=120uA so R1=10KΩ
Require VGB2=0.2, therefore R2=0.2/1.2*10 KΩ =1.67 KΩ [2 points]



Schematic with correct annotations [3 points]

[5]

4. This question relates to cochlear implants for restoring hearing.

a) Discuss the key limitations to state-of-the-art implantable cochlear prostheses [5]

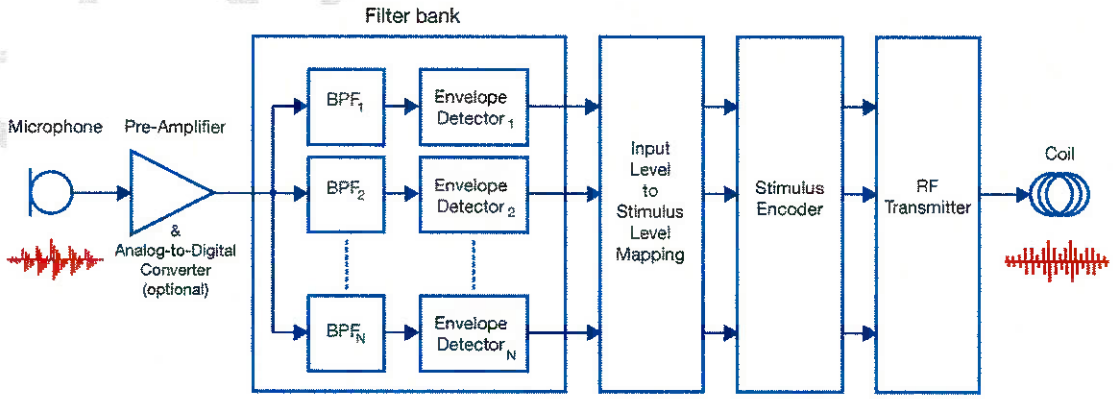
Maximum number of electrodes (currently around 20) is limited by current spreading due to fact they are implanted in a highly conductive medium (fluid) [1]. This limits the spectral resolution (20,000 hair cells being replaced by 20 electrodes – very coarse interfacing) [1]. Also, due to implantation of electrode array not all electrodes may be active/appropriately spaced so in most cases even fewer are being addressed. [1] Due to constraints in stimulation strategy the timing/phase information of the audio signal is not encoded/preserved [1]. This results in poor performance of CIs when it comes to music and/or directional hearing (in bilateral implants (1).

b) Compressed analogue (CA), Spectral Peak (SPeak) and Continuous Interleave Sampling (CIS) are three stimulation strategies that have been developed for cochlear implants. Describe in general what is the purpose of a stimulation strategy, and compare those stated above. [5]

Stimulation strategy mitigates the challenge that for electrical stimulation, if more than one channel is active at one moment in time there is current spreading/crosstalk. [1] In biology naturally, for example in the cochlear, the signals are communicated via the hair cells in parallel (i.e. all 20,000 cells operate concurrently). [1] In the case of electrical neural stimulation we can't do this- so need a strategy as to how we can effectively time division multiplex (e.g. CIS) in the most effective way as to achieve the desired perception (e.g. understanding speech) [1]. It is also possible through advanced stimulation strategies to implement multipolar electrode configurations aiming to focus or direct the stimulation [1]. Finally, we can use known characteristics of the signal to prioritse which channels get stimulated first (or to select a subset), such as what is done in SPeak. [1]

c) Sketch a functional block diagram of a typical cochlear implant, briefly annotating what is the function of each block. [5]

Key features that need to be annotated: Front end (microphone, amplifier, data converter) [1], audio processing (filters, power detection, compression) [1], stimulation strategy (sequencing/selection/and level mapping) [1], stimulation circuit and electrode array [1], transcutaneous telemetry (need to communicate power/data across skin) – and it contains implanted and external modules [1]. See example figure below.



a) Given a cochlear implant uses two Zinc Air batteries (each with 600mAh capacity @ 1.4V) and that the stimulator requires an average stimulation current of 200μA @ 15V, estimate the battery lifetime of the entire system stating any assumptions made. [5]

- Two batteries connected in series to give 2.8V supply. [0.5] Converting from 2.8V supply to 20V supply: at efficiency ~50%. [1]
- So current needed from battery to produce 200uA at 15V = 200uA x 15/2.8/0.75 = 2.14mA at 2.8V [1]
- Then assuming this is half the total power consumption (including for other blocks) = ~5.3mA from 2.8V supply. [1]
- Finally given we have a 600mAh battery (and we are using both in series), then the battery lifetime would be 600mAh/4.3mA = 140 hours. [1]
- Assuming this is used ~16 hours per day -> ~8 days lifetime. [0.5]

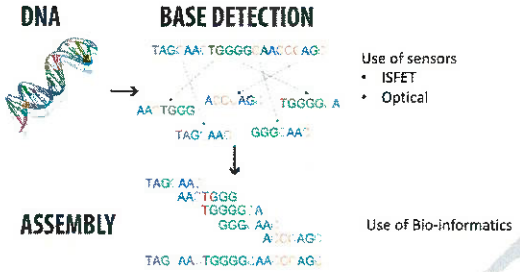
SOLUTIONS

5. This question relates to the design of an instrument for conducting DNA sequencing.

a) Explain the process of DNA sequencing describing the end-to-end procedure from an unknown strand of human DNA and the sensing mechanism used.

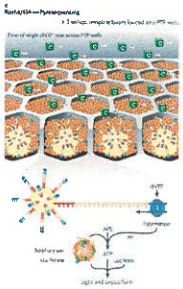
Take the human genome under investigation.
Chop the 3.2 Billion base pairs into strands of 100-1000 bases with overlaps. These are called fragments.
Immobilise the primers on magnetic beads.
Put each beads in a well on top of a sensor.
Detect the bases of the fragments by a method such as pyrosequencing.
Put the puzzle back together by using an assembly algorithm which detects overlaps in primer DNA.
Then describe the slides below for each of the points mentioned above.

DNA detection and assembly



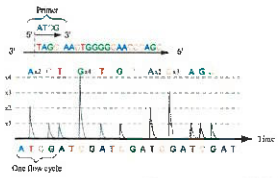
DNA sequencing technology

- Attach DNA fragments to magnetic beads.
- Magnetic beads sit in wells on top of sensors.
- ISFET sensors are being used in next generation sequencing technology.
- A base pair match causes a change in pH which we can detect.



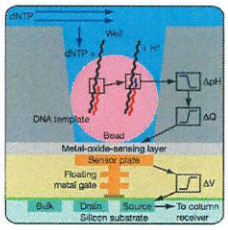
DNA sequencing technology

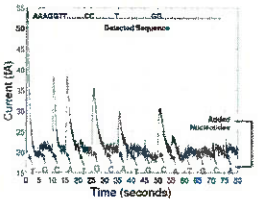
- Pyrosequencing
- Insert bases sequentially and see if there is a match.



Semiconductor Sequencing

- Use an ISFET array to detect a change in pH when there is a match.





[Total 10 points]

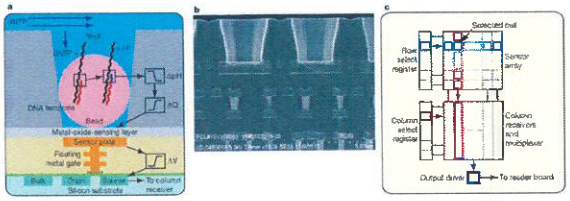
b) DNA sequencing has been revolutionised with the introduction of semiconductor sequencing. This uses a CMOS chip with millions of chemical sensors to conduct an in-parallel investigation of the DNA. You are required to design this CMOS chip describing the following:

- The choice of your sensor to detect DNA and the mechanism of sensing.
- The sensors operating characteristics, showing equations describing its function.
- The choice of instrumentation for this sensor.
- The circuit diagram implementation for an array to scale the design to millions of sensors.

Student needs to identify that this is an ISFET and explain its operation, show the method of instrumentation with justification and then sketch the circuit diagram of a 2x2 array taught in the lecture.

ISFETs for Genetic Sequencing

- Sensor and chip architecture



pH sensitive ISFETs

- Threshold voltage shifts with pH concenration.
- 1 pH unit causes a change of 59mV.
- This is known as the Nernst Potential - 59mV/pH

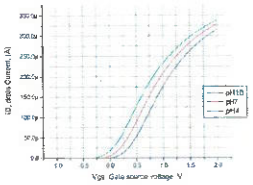
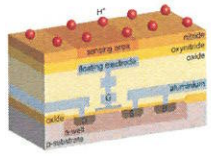
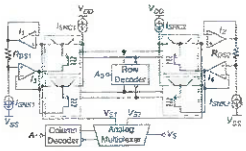


Figure 4.2: Measured I_D-V_{GS} characteristic for variable pH.

Large Scale Integration

- ISFETs are inherently CMOS compatible and can be used to make large scale sensing arrays.
- Scaling is dictated by Moore's law just like the MOSFET.





Equivalent Macromodel

Linear Region: ($V_{DS}<V_{GS}-V_t$)

Drain Current: $I_{DS(ir)} = 2\beta(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}$

Saturation Region: ($V_{DS}>V_{GS}-V_t$)

Drain Current: $I_{DS(sr)} = \beta(V_{GS} - V_T)^2(1 + \lambda V_{DS})$

Transconductance: $g_{m(sr)} = 2\sqrt{\beta \cdot I_D}$

Current Factor: $\beta = \frac{KW}{2L}$

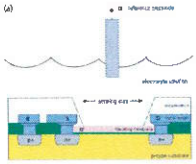
ISFET Threshold voltage: $V_T = V_{TH(MOS)} + V_{chem}$

Chemical Relation: $V_{chem} = \gamma + 2.3\alpha U_t pH$

γ is a grouping of chemical potentials, α is a sensitivity factor (0-1), U_t is the thermal voltage (26mV).

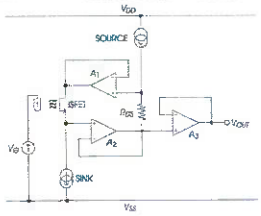


Figure 4.6: ISFET Behavioural Macromodel



ISFET readouts – Opamp based

- Constant charge method (CCM) is the most popular.
- Fix V_G, I_D, V_{DS} to operate at a fixed working point.
- V_S then tracks changes in V_T.
- Opamp A_2 tracks changes in VS and through feedback of A_1, sets V_D to ensure V_{DS} is always constant.



Circuit diagram of an ISFET source-and-drain follower circuit

[Total 10 points]

SOLUTIONS